Evidence-Guided Small-Part Sorting

A modular FANUC ER-4iA / Isaac Sim pipeline for camera-guided cube and screw sorting, with measured tests of neural perception and AI grasp ranking.

MANUFACTURING PROBLEM

Small-part sorting becomes unreliable when parts arrive at varying positions, overlap, or are partly occluded. A failed pickup adds recovery time, operator intervention, and lost throughput. We prioritize **correct placement and consistency**: a faster attempt has little value when it creates a drop or retry.

PHYSICAL AI APPROACH

Rendered camera images are converted into structured object observations and robot-base poses. The system generates multiple grasps, rejects candidates that violate reachability, finger fit, or clearance, selects a grasp, executes it in Isaac Sim, verifies final bin containment, and logs failures. Simulator truth is evaluator-only, never a runtime input.

HYPOTHESIS AND COMPARISON

Would pretrained AI improve perception or grasp selection? We compared deterministic geometry with a fine-tuned YOLO11n detector and NVIDIA Cosmos3-Nano, a pretrained physical-reasoning model used to rank already-feasible candidates. Cosmos was not trained by our team, and its confidence was not treated as a calibrated success probability.

SCOPE AND ASSUMPTIONS

Rigid, durable cubes and isolated lying M6x30 screws; bounded flat workspace; known part families, dimensions, and destinations; adequate visibility; destination capacity managed externally. Arbitrary piles, deformable parts, reflective finishes, and full bins remain outside the validated scope.

MATCHED ISAAC SIM PILOT

Ten preregistered randomized scenes (1-10 objects) were replayed under four policies: 40 end-to-end runs and 55 object opportunities per policy. Lighting, materials, camera, sensor effects, object poses, and physics varied. All reports completed with zero missing or episode-level infrastructure failures.

Policy	Sorted	Scenes	/ sim min
DD	**45/55 (81.8%)**	**6/10**	**4.97**
DA	38/55 (69.1%)	3/10	3.76
YD	14/55 (25.5%)	1/10	4.28
YA	9/55 (16.4%)	1/10	3.09

DD = deterministic vision + deterministic grasp; **DA** = deterministic vision + guarded Cosmos; **YD** = YOLO-assisted vision + deterministic grasp; **YA** = both.

EVIDENCE-BASED DECISION

DD exceeded DA by 15.2 percentage points in matched object completion (paired 95% CI: 3.3-29.9 points). Cosmos changed the deterministic grasp 21 times; only 9 changed choices succeeded. The strict YOLO integration rejected an entire observation when one detection disagreed with geometry. We therefore retain DD instead of forcing AI into the deployed path.

SCALING TO PARTS AND GRIPPERS

For a new part, add CAD or measured dimensions, symmetry and destination rules; collect representative images; validate camera-to-robot pose; define compatible grasps; then test on held-out scenes. A new gripper requires updated finger geometry, stroke, TCP, collision envelopes, and candidate rules - not a full pipeline rewrite. Learning is enabled only after it beats the deterministic baseline.

CSI EQUIPMENT AND LIMITATION

We used the organizer-provided FANUC digital twin, robot/gripper geometry, screw CAD, and Isaac Sim physics. We did **not** receive an opportunity to test the final pipeline on CSI hardware. Physical calibration, controller timing, grip forces, clearance, and sim-to-real reliability remain unverified; these results are simulation evidence only.

REPOSITORY AND REPRODUCIBLE RESULTS

github.com/ALL-Applied-AI-Network/msoe-ai-club-rockwell-automation-nvidia-hackath-new-team-8-b27f0a8f50/tree/experiments/randomized-four-combo-v1